

EXPLAINED EXAMPLE 11 – OTTO CYCLE ANALYSIS

- ① The Mazda 3 SkyActiv engine has a volume of 0.002 m^3 and a compression ratio of 13. Assume the intake temperature is 300 K and intake pressure is 0.1 MPa, with a maximum cycle temperature of 1500 K. Calculate the net work, heat addition, and thermal efficiency of this engine if it runs an ideal air-standard Otto cycle. Use properties for air: $R=287 \text{ J/kg-K}$ and $\gamma=1.4$.

Given : $V_{\text{eng}} = 0.002 \text{ m}^3$
 ② $= V_1 - V_2$
 $r = \frac{V_1}{V_2} = 13$

$T_1 = 300 \text{ K}$

$P_1 = 0.1 \text{ MPa}$

$T_{\text{max}} = T_3 = 1500 \text{ K}$

③ Find : W_{net}
 Q_{in}
 η_{th}

Assume : ideal gas
 ② ideal cycle
 quasi-steady
 const. M

$\Rightarrow W_{\text{net}} = {}_1W_2 + {}_3W_4$
 $Q_{\text{in}} = {}_2Q_3$
 $\eta_{\text{th}} = \frac{W_{\text{net}}}{Q_{\text{in}}}$

} solve for T
 to solve for
 work and heat

→ State 1 : $T_1 = 300 \text{ K}$ $P_1 = 0.1 \text{ MPa}$

①

$V_1 - V_2 = 0.002 \text{ m}^3$
 $V_1 = 13V_2$

} $13V_2 - V_2 = 0.002 \text{ m}^3$
 $12V_2 = 0.002 \text{ m}^3$

$V_2 = 0.00017 \text{ m}^3$

$V_1 = 13V_2 = 0.00217 \text{ m}^3$

② : $P_1 V_1 = M R T_1 \rightarrow M = \frac{P_1 V_1}{R T_1} = \frac{(100000)(0.00217)}{(287)(300)}$

$M = 0.0025 \text{ kg}$

→ State 2 : $P_2 v_2 = MR T_2$

①

$$T_2 = \frac{P_2 v_2}{MR} \Rightarrow P_2 = P_1 r^\gamma$$

$$= 100000 (13)^{1.4}$$

$$= 3626775.7 \text{ Pa}$$

$$= \frac{(3626775.7)(0.00017)}{(0.0025)(287)} \rightarrow \boxed{T_2 = 859.3 \text{ K}}$$

② → State 3 : $\boxed{T_3 = 1500 \text{ K}}$

① → State 4 : $T_4 = T_3 \left(\frac{1}{r}\right)^{\gamma-1} = 1500 \left(\frac{1}{13}\right)^{0.4} \Rightarrow \boxed{T_4 = 537.7 \text{ K}}$

⇒ $W_{\text{net}} = {}_1\bar{W}_2 + {}_3\bar{W}_4 \Rightarrow \text{First law: } M \Delta u = \cancel{Q} - \bar{W}$

② → ${}_1\bar{W}_2 = -M c_v (T_2 - T_1)$ where $R = c_p - c_v$

$$= -(0.0025)(717.5)(859.3 - 300)$$

$$= -1003.24 \text{ J}$$

$$\gamma = c_p / c_v$$

$$R = \gamma c_v - c_v$$

$$c_v = \frac{R}{\gamma - 1} = \frac{287}{0.4} = 717.5 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

① → ${}_3\bar{W}_4 = -M c_v (T_4 - T_3)$

$$= -(0.0025)(717.5)(537.7 - 1500)$$

$$= 1726.13 \text{ J}$$

⇒ $W_{\text{net}} = {}_1\bar{W}_2 + {}_3\bar{W}_4 \rightarrow \boxed{W_{\text{net}} = 772.89 \text{ J}}$

$$\Rightarrow \textcircled{1} \quad Q_{in} = {}_2Q_3 \rightarrow \text{first law: } M\Delta u = Q - \bar{W}$$

where ${}_2\bar{W}_3 = \int_{T_2}^{T_3} P dV = 0$ b/c $\Delta V = 0$

$$\Rightarrow {}_2Q_3 = M C_V (T_3 - T_2)$$

$$= (0.0025)(77.5)(1500 - 859.3)$$

$$\boxed{{}_2Q_3 = 1149.26 \text{ J}}$$

$$\Rightarrow \textcircled{2} \quad \eta_{th} = \frac{\bar{W}_{net}}{Q_{in}} = \frac{772.89}{1149.26} \rightarrow \boxed{\eta_{th} = 0.673}$$